Each of these is meaningful and can be tested by orienting $\mathcal{A}$ along the appropriate axis. The negation of each is also meaningful. For example, the negation of the first proposition is

> **not** A: The *z* component of the spin is −1.

But now consider the composite propositions

> (*A* **or** *B*): The *z* component of the spin is +1 or the *x* component of the spin is +1.
>
> (*A* **and** *B*): The *z* component of the spin is +1 **and** the *x* component of the spin is +1.

Consider how we would test the proposition (*A* **or** *B*). If spins behaved classically (and of course they don't), we would proceed as follows:[5]

- Gently measure σ_z and record the value. If it is +1, we are finished: the proposition (*A* **or** *B*) is true. If σ_z is −1, continue to the next step.

- Gently measure σ_x. If it is +1, then the proposition (*A* **or** *B*) is true. If not, this means that neither σ_z nor σ_x was equal to +1, and (*A* **or** *B*) is false.

There is an alternative procedure, which is to interchange the order of the two measurements. To emphasize this reversal of ordering, we'll call the new procedure (*B* **or** *A*):

- Gently measure σ_x and record the value. If it is +1 we are finished: The proposition (*B* **or** *A*) is true. If σ_x is −1 continue to the next step.

- Gently measure σ_z. If it is +1, then (*B* **or** *A*) is true. If not, it means that neither σ_x nor σ_z was equal to +1, and (*B* **or** *A*) is false.

In classical physics, the two orders of operation give the same answer. The reason for this is that measurements can be arbitrarily gentle—so gentle that